1 The nature of technology: Digital devices

You will learn

- what a digital device is
- the digital devices you can use
- how computers can help you
- things computers cannot do.



In this unit you will learn about digital devices. Devices are things people make to help them. Digital devices have got a computer inside them. The computer makes the devices even more useful.

Did you know?

A computer can check your heartbeat.
The computer uses safe electrical sensors.
A doctor looks at the computer screen.
The doctor can tell if your heart is healthy.



Talk about...

What job would you like to do when you grow up? Will you use computers in the job? How will computers help you?



Learning outcomes: Describe a range of familiar digital devices; Describe tasks where computers can be helpful



Class activation

Some people have jobs that help you. For example, a doctor helps people get better if they are ill.

What other jobs can you think of that help people? Make a class list of these jobs. Next to each job say one way the person might use a computer.

digital device
processor input device
output device touchscreen
hand-held mobile device
tablet computer smartphone





1.1 Digital devices

In this lesson

You will learn:

- what digital means
- what devices are digital.

Spiral back

Last year you learned that the computer uses electricity. In this lesson you will find out more about how a computer uses electricity. You will also learn why a computer is called a digital device.

Switches

A computer works by electricity. Inside the computer there are millions of electrical switches. They are not switches like the light switch. You cannot see them or touch them. They are tiny switches that work by electricity.

The switches inside the computer can be on or off.

- You show an off switch as the number 0.
- You show an on switch as the number 1.

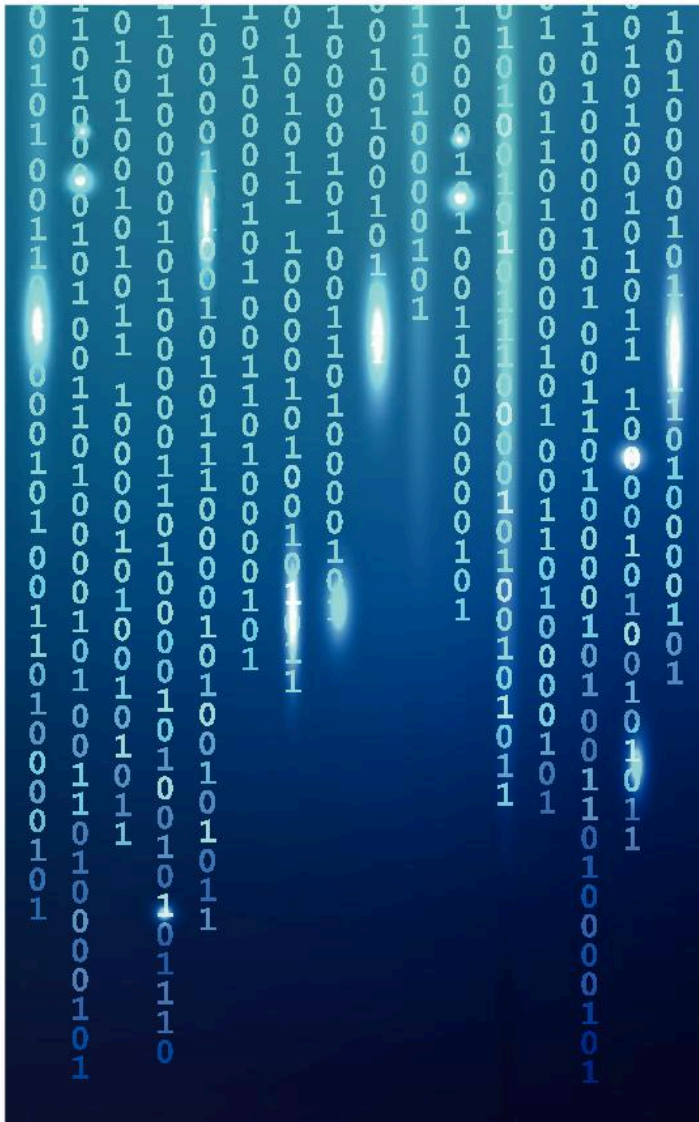
The switches are always changing. Electricity goes through the computer. The flow of electricity turns the switches on and off.

Digital

Computers use the on-off switches to make electronic numbers. That is all there is inside a computer. There are millions and millions of electronic numbers.

Everything inside a computer is made out of those numbers.

Digital means made out of numbers. Everything inside a computer is digital.



Digital device

A device is anything that people make that is useful and helpful.
A **digital device** is any device that has a computer in it.

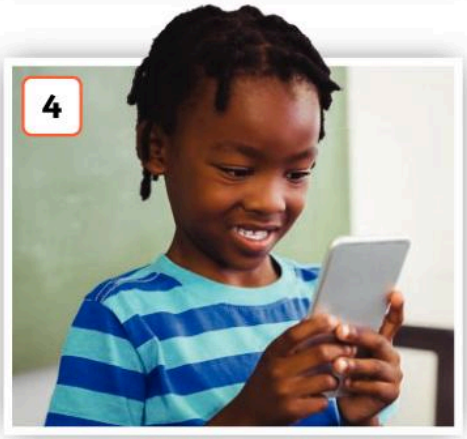
There are many types of digital device. Which ones do you know?

Smartphone | Desktop computer | Tablet | Laptop computer











Activity

Match the devices to the names. Write down the names and numbers.
Or draw the devices and write the names next to the drawing.



Extra challenge

Another type of digital device is called a wearable device. Have you heard of that before? Find out more. Draw or write to show what you found.

Think again

Say something that is not a digital device. Explain your answer.

1.2 The parts of a computer

In this lesson

You will learn:

→ about the parts of a typical computer.

Spiral back



Last year you learned about input and output devices. In this lesson you will learn how they make a computer system.

Processor

Inside every computer is a **processor**. A processor is an electronic device. The processor controls all the other parts of the computer.

Input and output

Other devices are attached to the processor:

- **Input devices** send signals to the processor. Examples of input devices are the mouse and keyboard.
- **Output devices** get signals from the processor. Examples of output devices are the screen and speaker.

The input and output devices let you use the computer.

Desktop computer

In a desktop computer all the devices are quite big. For example, the screen is big. There might be more than one screen.

The devices are separate from each other. They might be joined by wires. They might be wireless.

A desktop computer is big and heavy. It is difficult to carry.



Laptop computer

A laptop computer has smaller parts. The screen and the keyboard are small. All the parts are held in the same case. They are fixed in place.

A laptop is small and light. It is easy to carry.

The touchpad is used instead of a mouse. Some people take a mouse with them and plug it into the laptop.

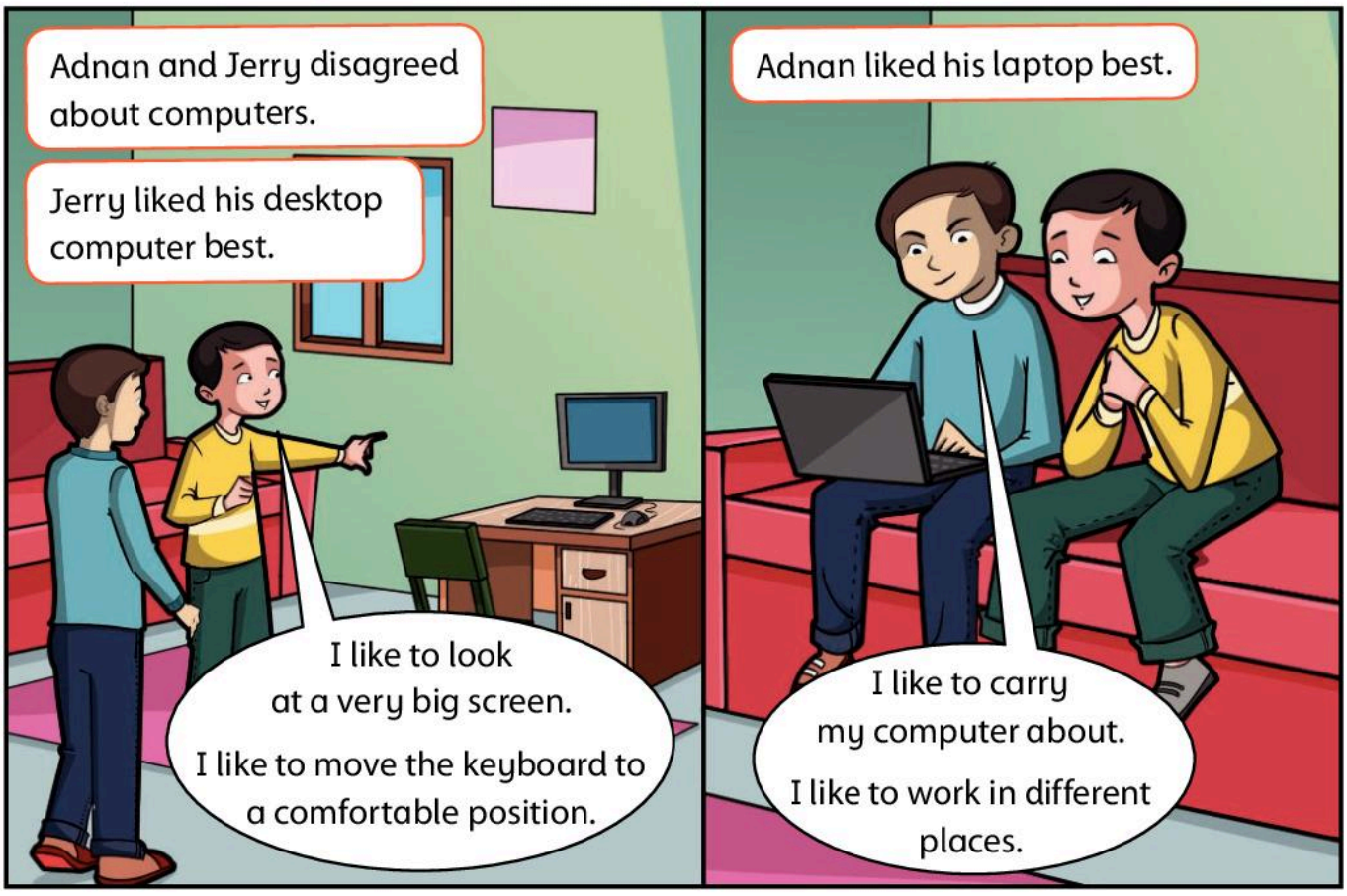


What is a desktop computer used for?

You can use a desktop computer when you want to do lots of work. You can sit and do a lot of typing. You can see your work clearly on a big screen.

Many people like to play games on desktop computers. They like the big screen. They like to use the keyboard and other controls that plug in to the computer.

Which would you choose?



Activity

What type of computer do you use most at school? Draw the computer and label the parts.

Extra challenge

Which do you like best – a desktop or a laptop? Make an advert for the one you prefer.

Think again

What device is at the centre of a computer system? (**Hint:** It is not an input or output device.)

1.3 Mobile devices

In this lesson

You will learn:

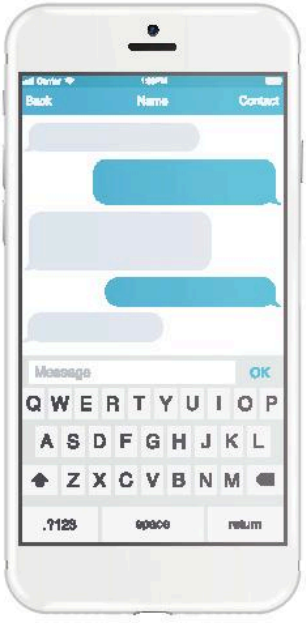
➔ about the features of mobile devices.



Touchscreen

Some computers have a **touchscreen**. A touchscreen is used for input and output.

- The screen shows you pictures, words and video.
- You can touch the screen to tell the computer what to do.



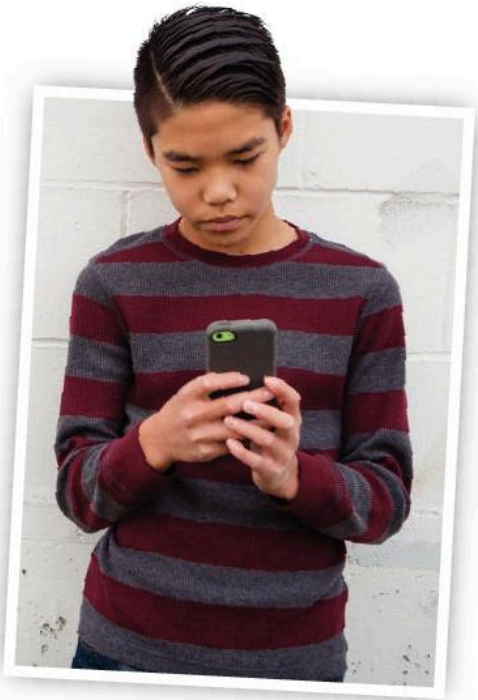
If you have a touchscreen, you do not need a keyboard. A picture of a keyboard appears on the screen. You can touch the screen to use the keys.

Hand-held computers

A computer with a touchscreen can be very small and light. You can hold it and use it in your hands. You can carry it with you wherever you go. Computers like this are called **hand-held** computers. Another name is a **mobile device**.

A **tablet computer** has quite a big screen. It is about as big as a book.

A **smartphone** is a mobile phone with a computer inside. It is much smaller than a tablet.





What are mobile devices used for?

People use mobile devices to look at the internet. Mobile devices have a wireless connection.

- A tablet uses the wireless network in a building. For example, there may be a wireless network in your school or home.
- A smartphone has its own internet connection. You can use it outdoors, in the country or the town.

People use mobile devices mainly to look and read. They might watch a TV show or read a website. They might type a short message to a friend. They might type quick notes.

People do not usually use mobile devices to make a big document with lots of typing.



Activity

Write a list of all the digital devices you can see in this picture.





Extra challenge

Choose one of the people in the picture. Write a paragraph about:

- what type of digital device they are using
- what they might be doing with it.

You can write about anything you like.

Think again

What are the differences between a smartphone and a tablet?

1.4 Computers at work

In this lesson

You will learn:

➔ how computers are used in different jobs.

The doctor



Doctors help you when you are sick. They help you to get better.

Computers can help the doctor do her work. For example, the doctor can use the computer to:

- find out about diseases and treatments
- send messages to doctors in other hospitals
- make appointments for patients
- look inside the human body to take accurate pictures and measurements
- print out charts and diagrams to help spot illness
- store facts about patients
- control machines to care for a patient day and night.

The shopkeeper



Shopkeepers help you. They help you buy the things you need.

Computers can help the shopkeeper do his work. For example, the shopkeeper can use the computer to:

- sell things on the internet to people all over the world
- find out about new products
- store facts about the items for sale

- add up your bill quickly and accurately
- make colourful posters and leaflets to advertise
- take payments from customers
- pay wages to staff
- store food at an even temperature at all times.



Activity

Think of a job you would like to do when you become an adult. Think of one way you might use a computer in that job. Draw and write to show how you would use a computer.











Extra challenge

You may not use a computer in all parts of the job. Draw and write to show the things you would do without a computer.

Think again

Talk to an adult who has a job, for example, someone in your family. Ask them if they use a computer. Find out how the computer helps them.

1.5 How computers help

In this lesson

You will learn:

- about the benefits of using a computer.

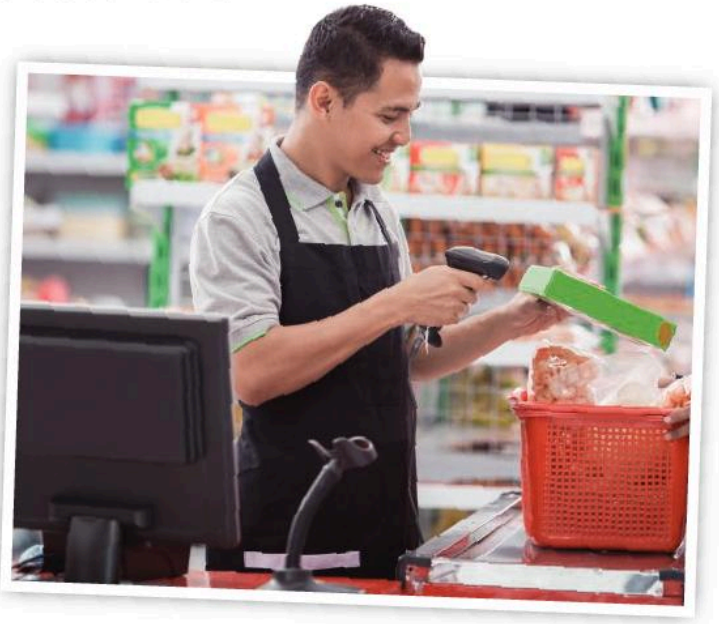


Benefits

'Benefit' means to make things better. People use computers because they bring benefits. Computers can help you to do things better.

Here are some benefits of using computers:

- **Make work easier.** The computer can make it easier to do some tasks.
- **Fast and accurate.** The computer does accurate sums and measurements.
- **Store facts and find things out.** The computer can store facts and find answers, including using the internet.
- **Long-distance communication.** The computer can send messages to people all over the world.
- **Clear and neat work.** The computer makes output that is neat and clear.
- **Work without rest.** The computer can control machines and devices day and night without rest.



In the last lesson you learned about some ways the computer helped a doctor and a shopkeeper.

The table on the next page shows these benefits for a doctor and a shopkeeper. There are some gaps.

Benefits	Doctor's work	Shopkeeper's work
Make work easier	Make appointments for patients.	Take payments from customers. Pay wages to staff.
Fast and accurate		
Store facts and find things out		Find out about new products. Store facts about the items for sale.
Long-distance communication	Send messages to doctors in other hospitals.	
Clear and neat work	Print out charts and diagrams to help spot illness.	
Work without rest		Store food at an even temperature at all times.



Activity

Make a copy of the table. Look back at the last lesson. Look at all the tasks a doctor and a shopkeeper do. Put each task into the right box of the table. Some have already been done for you.





Extra challenge

You have looked at the job of a doctor and a shopkeeper. Now think of the job of a teacher. Think of all the ways a teacher might use a computer.

- Write down all the benefits you can think of.
- Can you link them to the benefits on this page?
- Make a final column for the table to show the benefits to a teacher.



Be creative

Choose one of the jobs shown on this page (doctor, shopkeeper or teacher). Instead of words, draw cartoons to show how they would use computers.

Think again

Give examples from your school work where the computer has helped you:

- to find out facts
- to do clear and neat work.

1.6 Making good choices

In this lesson

You will learn:

- how to choose tasks where a computer can or cannot help
- how to choose the right computer for a task.



Limitations

Computers have limitations. That means things the computer is not so good at.

Here are some things that computers cannot do.

- **Do something creative.** The computer cannot invent new things or be creative.
- **Understand people.** The computer cannot understand how people feel. The computer cannot show human feelings or make friendships.
- **Take responsibility.** Computers can give people information to help them make decisions. But you must make your own choices.

Limitations	Doctor's work	Shopkeeper's work
Doing something creative		
Understanding people		
Taking responsibility		

Here are some examples of tasks that a doctor or a shopkeeper might do.

Decide between two treatments.

Listen to a patient talk about their worries.

Talk to a customer about what they want to buy.

Create an advert for the shop.

Decide to change the prices in the shop.

Invent a new way to cure a disease.



Activity

Make a copy of the table on the previous page. Put the activities into the table to show things a doctor or shopkeeper can do that a computer cannot do.

Choose the right device

If you use a computer for a task you need to choose the right device. Here are the devices you have learned about in this unit.

Desktop computer	
Laptop computer	
Tablet	
Smartphone	



Activity

Here are some things people said. Recommend the ideal digital device for each person.

- "I want to do a lot of typing sitting at my desk."
- "I want to play a computer game on the bus to school."



Extra challenge

Pick a digital device you did not choose. Say one thing a person might do with that device.

Think again

Say one thing a teacher can do that a computer cannot do.



Digital citizen of the future

At school you often have to use a computer because your teacher says so! But when you are grown up you will be able to make decisions.

- Do you want to use a computer?
- What device do you choose?

Learning about computers at school will help you make good decisions in the future.

Check what you know

You have learned

- what a digital device is
- the digital devices you can use
- how computers can help you
- things computers cannot do.

Test

1. Say one task you have done at school using a computer.
2. How did the computer help you do the task?
3. An artist used a computer to help with his work. Explain two ways a computer could help him.
4. Explain some parts of an artist's job that a computer could not help with.



Activities



1 What device are the boys using? Draw a picture of the device. Show it as clearly as you can.

2 One part of the device is used for input and output. Label this part.

3 Think about how you use the device.

- Is the device easy to carry about from place to place?
- Would you use this device to type for 1 hour?

4 Say one way you might use the device in a future job.

Self-evaluation

- I answered test questions 1 and 2.
- I completed activity 1. I drew a picture of a device.
- I answered test questions 1–3.
- I completed activities 1 and 2. I drew a picture of a device and added a label.
- I answered all the test questions.
- I completed all the activities.

Re-read any parts of the unit you feel unsure about. Try the test and activities again – can you do more this time?